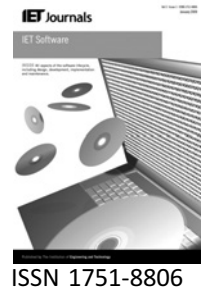


Published in IET Software
 Received on 14th May 2009
 Revised on 11th December 2009
 doi: 10.1049/iet-sen.2009.0045



Using a web-based tool to define and implement software process improvement initiatives in a small industrial setting

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Abstract: Top-down process improvement approaches provide a high-level model of what the process of a software development organisation should be. Such models are based on the consensus of a designated working group on how software should be developed or maintained. They are very useful in that they provide general guidelines on where to start improving, and in which order, to people who do not know how to do it. However, the majority of models have only worked in scenarios within large companies. The authors aim to help small software development organisations adopt an iterative approach by providing a process improvement web-based tool. This study presents research into a proposal which states that a small organisation may use this tool to assess and improve their software process, identifying and implementing a set of agile project management practices that can be strengthened using the CMMI-DEV 1.2 model as reference.

1 Introduction

Nowadays, software processing in an industrial environment is key to the productivity and efficiency of development activities and competitive strategies against adversaries. However, since the beginning of the 1980s the software industry has tried to increase its quality and productivity by applying new methods and techniques. It has been recognised that unfortunately the fundamental problem for many companies is the incapacity to manage the software process. In recent years, there has been on-going demand for better services and functionality in software products. Many methods, techniques and tools have been developed. Nevertheless, software products still suffer from excessive costs, delays in delivery and low quality.

According to the Prosoft Foundation (Program for Develop the Software Process) [1] and researchers such as Oktaba [2] and Brodman [3], the software development process is far from being a 'mature process'. To satisfy the various requirements for software processing both large companies and small and medium size enterprises (SMEs)

have made a central effort in software process improvement (SPI). As a result, SPI has emerged as a new way of solving these problems. An indicator of this is the increasing number of international initiatives related to SPI, such as capability maturity model integration for development v1.2 (CMMI-DEV) [4], ISO/IEC 15504:2004 [5], SPICE [6] and ISO/IEC 12207:2004 [7]. These SPI methods have been presented as an alternative to achieving an increase in the quality of services and products that the software organisation provides. These trends in the software industry have emerged since recognising that the absence of software process management causes general problems during software development, and being aware of the limitations on the introduction of tools and technology. In addition, many methods for evaluating improvements in organisations, such as the standard CMMI appraisal method for process improvement (SCAMPI) [8], ISO/IEC 15504:2004 [5] and capability maturity model-based assessment for internal process improvement (CBA-IPI) [9] and improvement models such as IDEAL [10] have been developed.

This interest in software improvement in large enterprises is now being extended to small and medium settings. However, the problem is the high implementation cost, which is independent of the size of the company [11]. Because models have been developed for large enterprises, only a few small and medium software organisations are aware of them. It is recognised that for small settings, projects or teams, process improvement efforts are difficult and frequently overwhelming; however, the need for improvement remains and cannot be ignored [12]. For example, the Mexican Government financed the SPI initiative through the Economics Secretary with the ProSoft Program to guide and support the use of effective practices in industrial small settings [1]. This kind of financing, of which similar examples can be found in most countries around the world [13, 14], attempts to demonstrate that with the adoption of best software practices, the range of advantages that small organisations can hope to receive could be expanded; and the use of methods and technologies available to software development could be significantly improved according to the organisational needs. In spite of this, SPI is not explored sufficiently by IT departments in many organisations; hence any development of an SPI initiative becomes completely chaotic as important concepts are unknown or not applied.

Apparently, SMEs typical issues without software process management activities are as follows; no principles for the manner of performing tasks including plans and procedures, ad hoc task performance relying only upon the experience of practitioners and an insufficient number of staff with appropriate skills, and more. However, many studies [15, 16] have shown that the Capability Maturity Model Integration (CMMI) [17] based process improvement can enable small organisations to more consistently deliver better products and services on time, with higher quality, and at the predicted costs. Process improvement initiatives are impacted by many factors (e.g. financial resources, human resources, business objectives, culture of the organisation and vision).

When starting a process improvement initiative, it is imperative to determine the appropriate scope and tasking, taking into account the size of the organisation. More small organisations are initiating SPI efforts. Once they are committed to starting this process, the next step is the assessment of the current capability of the organisation process to develop and maintain software. Following this, they have to define the action plans related to the implementation of the selected processes to be improved. There is a large quantity of case studies that demonstrate the applicability of these approaches in a small business environment (e.g. studies published in [18], research by Cater-Steel [19] and Garcia *et al.* [20]); however, unfortunately there is a lack of information on software tools that actually guide small organisations to conduct the process improvement programme. To handle these issues

and avoid a worst-case scenario, more viable approaches for implementing, measuring and improving small organisations' software processes are strongly recommended. In other words, it is necessary to conduct software process assessment and improvement as a matter of primary concern. Furthermore, small software development organisation needs an automated tool support for the way it effectively manages its software processes [21].

1.1 CMMI for development

According to SEI, 'CMMI is a process improvement maturity model for the development of products and services. It consists of best practices that address development and maintenance activities that cover the product lifecycle from conception through delivery and maintenance. This latest iteration of the model as represented herein integrates bodies of knowledge that are essential for development and maintenance. These, however, have been addressed separately in the past, such as software engineering, systems engineering, hardware and design engineering, the engineering "-ilities", and acquisition' [17]. The prior designations of CMMI for systems engineering and software engineering (CMMI-SE/SW) are superseded by the title 'CMMI for development' to truly reflect the comprehensive integration of these bodies of knowledge and the application of the model within the organisation. CMMI-DEV 1.2 provides a comprehensive integrated solution for development and maintenance activities applied to products and services.

The CMMI-DEV official report indicates that: 'CMMI for Development V1.2 is a continuation and update of CMMI V1.1 and has been facilitated by the concept of CMMI "constellations" wherein a set of core components can be augmented by additional material to provide application-specific models with highly common content. CMMI-DEV is the first of such constellations and represents the development area of interest'. [The differences between v1.2 and v1.1 are explained in detail on the SEI web site (www.sei.cmu.edu) in CMU/SEI 2006 TR-008.]

To improve software development practices, practitioners, projects and organisations must move from ad hoc practices to explicit software development practices. Using CMMI-DEV and the IDEAL model [10], a simple improvement process, organisations can do just that. The IDEAL model is used as the approach for this improvement process. The steps of the IDEAL model are outlined below:

- *Initiating*: Laying the groundwork for a successful improvement effort.
- *Diagnostic*: Determining where you are relative to where you want to be.
- *Establishing*: Planning the specifics of how you will reach your destination.

- *Acting*: Doing the work according to the plan.
- *Learning*: Learning from experience and improving your ability to adopt new technologies in the future.

Using the IDEAL model and CMMI-DEV, a process improvement team would follow each phase in the loop to improve its organisation's software development practices.

2 Related work

We have researched the characteristics of typical SPI tools for small software development organisations and projects. Unfortunately, there is not too much research about SPI tools development to help the improvement initiatives in SMEs. SPI tools generally automate the process of assessment based on the standard model framework for software processes, such as ISO/IEC 15504:2004 [6] and CMMI [17]. The existing SPI tools usually include some or all of the following characteristics [21–23]:

- Collecting and managing general company information, assessment goals, project and process instance information, assessment items and so on.
- Selecting which processes to assess.
- Generating questionnaires in accordance with various types of software process standards.
- Modifying the software process model for domain adaptation and the changed versions of the process model.
- Self-evaluating tasks by software practitioners.

However, the main objective of these tools is to conduct the process assessment to ensure a project's compliance with specific standards in software development organisation. Thereby they can be intended only to automate the process of software process assessment. All software tools for implementing and maintaining improvement initiatives can be divided according to the model under maintenance: CMMI [17] or ISO/IEC 15504 [6]. The other aspect of division is open source software (including academic tools) and commercial software. According to [24] analysing open source software, attention can be drawn to the fact that the better known tools are developed for the models of the CMM family. Beyond any doubt there exists some non-commercial tools for ISO/IEC 15504, but in most cases this is exploratory software (The exploratory software term refers to software created by Universities Labs around the world only with research objectives. It means only software created to explore and analyse new alternatives of assessment and improvement within organisations.] created by research objectives within Universities around the world. The need for tools, enabling efficient and reliable capture of

assessment data are addressed in the SPICE project through notion of an assessment instrument.

Research by He *et al.* [25] stated that 'quality improvement and management practices involve using tools, which were formulated from industrial practices and quality engineering theory'. They point out that the seven old Japanese tools (p.e. check sheet, histogram, pareto analysis) claim to be able to solve up to 95% of quality-related problems and that the other seven Japanese tools (p.e. affinity diagrams, relation diagram, tree diagram) claim to have started a new era for quality. However, they comment that these tools had not been discussed systematically in the context of the software sector.

Since then, a number of tools have emerged to support the data collection and storage procedures, for example research by Him Lok and Walker [26] and Hunter *et al.* [27]. In [26] authors described the development of an automated tool to support the conduct of assessments in Phase 2 of the trialing period using the embedded model described in Part 5 of ISO/IEC 15504 [6], the SEAL of Quality SPICE assessment tool (SEAL OQ). The development process and product quality objectives and their validation were described.

Other researchers investigated effective ways to visualise acquired assessment data. The tool software by Robin Hunter [28] from Strathclyde University is an example. This software was created using MS Access and its aim was to explore software process visualisation. The demand for import/export tools emerged after it was noted that Office programs (such as Excel and Access) were useful in analysing the information elements of the model under development; the development of additional tools was therefore not necessary. Furthermore, project developers and assessors mastered the Office programs; it is more convenient to perform a partial analysis using Office programs. A fair amount of open source software was developed by the assessors and is possible to adapt to the Office programs; in fact all material could be exported.

The research work of [29] presented an approach to build a computational environment to support SPI activities that used the process-improvement activity support system (PIASS). PIASS [30] was developed as an integrated environment that provided tools necessary to make an assessment of targeted software processes, and a knowledge-base to store guidelines, experiences and know-how that were necessary to proceed with SPI activities for a particular organisation.

Research by Fukuyama *et al.* [31] described a software process improvement support system (SPIS) that indicated 'how to' perform SPI activities. It was designed to be used mainly for internal SPI activities by software development companies and their project teams. The SPIS was composed of seven support tools and promotes SPI

activities. The tools included a specific capability maturity model, rules for assessing a company's maturity level and an assessor qualification system for organising the assessment team. These tools were extracted deductively from the researchers experience and had proved to be effective when used for the SPI activities at one large company.

The tool concept proposed in [32] was based on the seamless integration of different measures tools into a flexible, extensible and efficient quality assessment structure. This setup allowed authors to assess the quality of a software system in real-time and paves the ground to establish continuous quality controlling progress.

Choi *et al.* [33] suggested that a process management system was needed in order to ultimately assist SPI activities. Authors proposed a system to provide two different approaches to its use. One was an internal assessment and improvement process at the level of software process appraisal. The other was a self-controlled improvement process at the level of the software practitioner to lead active participation in the enhancement of individual software processes. The approach of the proposed system was decided according to the group of system users.

The research work of [34] implemented SPI initiatives based on the Brazilian software process model (named MPS model) through the adoption of an SPI strategy (named SPI-KM [34]) and the support of a process-centered software engineering environment named Taba Workstation [35]. The Taba Workstation is a process-centered software engineering environment (PSEE) that supports software process definition, deployment and enactment. The software process definition related tasks were implemented through a CASE tool that used the ISO/IEC 12207 standard as a basis for defining software processes and also maturity models like MPS model.

Alagarsamy, Justus and Iyakutti considered the possibility of having a standardised knowledge management tool (KMT) that would facilitate SPI [36]. The authors have produced significant work [37, 38] that contributes to the research community and to the software engineering community in the areas of knowledge management and SPI.

The knowledge-based system proposed by Andrade *et al.* [22] evaluates an organisation at a determined CMM [39] level and as such limits the need for the services of an auditor in those cases in which the system's response complies with the requested CMM level and the necessary associated skills. This clearly implies an important cost reduction for audits with a negative result. The design of this system was based on the CommonKADS methodology [40], and its implementation was carried out with the Clips tool.

In analysing commercial software, most attention was paid to the commercial tools of one of the leading companies HM&S; these tools are designed for working both with

ISO/IEC 15504 [5] and with CMMI [17]. CMMI v1.2 Browser [41], for example, allows users to navigate easily through the CMMI-DEV. This CMMI browser covers the continuous as well as the staged representation. Wibas GmbH (for more information of Wibas GmbH visit www.wibas.com) develops a browser that can help organisations leverage the usage of CMMI. This company assumes that since the material in CMMI is for information purposes, it is most useful to look this information up when organisations need it.

CMM-Quest [42] is a HM&S IT-Consulting GmbH tool that allows organisations to perform self-assessment using the continuous representation of CMMI-SE/SW as a reference model. At first, the assessment was limited to assigning values to each objective and did not allow evaluations in specific and generic practice level (below objective values). Actually, the software tool rates and analyses the software development processes compliant to CMMI-DEV v1.2 and uses ISO/IEC 15504 as its assessment method. However, CMM-Quest is not easy to use.

Another commercial tool based on CMM-SE/SW is Appraisal Wizard [43] from Integrated System Diagnostic Inc.. Appraisal Wizard is based on the SCAMPI [8] method and supports almost all CMM models. This tool assigns all values to different assessment instances (practices, objectives, process areas) and provides large and detailed support; however, using Appraisal Wizard is also quite difficult because of the tool complexity.

SPICE 1-2-1 [44] is another software tool from SynSpace GmbH that directly assigns values to base and generic practices; it has a medium rate of difficulty when organisations try to use it. Also, SynEval [45] (another tool from SynSpace) helps the Assessor and/or particularly the Lead Assessor to argue around the results from several assessments to analyse them. The analysis mechanism of SPICE1-2-1 and SynEval works conform to ISO/IEC 15504 (SPICE).

Management Information Systems developed the free tool called IME Toolkit (interim maturity evaluation) [46] based on CMMI models. IME Toolkit assigns numerical values to practices and generates a score for process areas.

The MKS Integrity Suite [47] fully supports SPI initiatives and complies with certifications such as CMM and CMMI. According to MKS Inc. the MKS tool provides automated audit tracking and control enabling organisations to trace the origins and details of change to business critical applications, and even the documented software process itself.

Summarising, Table 1 shows a comparison among the analysed tools according to function and operation

Table 1 Software process assessment and improvement tools features

	Perform remote assessment?	Generate IPs?	Use an improvement tutor?	Works in small settings?	CMMI-DEV compliance	ISO/IEC 15504 compliance
SEAL OQ	no	no	no	no	no	yes
PIASS	yes	yes	no	no	no	no
SPIS	yes	no	no ^a	no	no	no
process management system	yes	yes	no	no	no	no
Taba workstation	no	no	no	yes	no	no
KMT	no	no	no	yes	yes	no
knowledge-based system	yes	no	no	no	no	no
CMMI v1.2 browser	no	no	no	no	yes	no
CMM quest	yes	no	no	no	yes ^b	yes
Appraisal Wizard	yes	no	no	no	no	no
SPICE 1-2-1	yes	no	no	no	no	yes
SynEval	yes	no	no	no	no	yes
IME Toolkit	yes	no	no	no	no	–
MKS Integrity Suite	yes	no	no	no	no	no

^aSPIS provides extensive guidelines, but not a formal tutorial

^bActually the software tool rates and analyses the software development processes compliant to CMMI-DEV v1.2

perspectives with the aim of highlighting some gaps and obtain an initial benchmark.

However, such commercial software is defined by a relatively large price tag (from \$900 to \$15 000 USD) which is out of the price range of SMEs. Its main features are a well organised subsystem of the assessment presentation, a sufficient amount of tools for maintaining the process improvement, and integration with other tools. However, despite all the advantages affirming the existence of efficient assessment and maintenance tools nowadays, there are also some considerable drawbacks. Furthermore, a few tools can really help a SME, taking into account its particular circumstances; but hardly provide an automated mechanism to create improvement plans (IPs). That is, there is a limit to being able to provide more helpful and diverse support for establishing SPI initiatives and managing the whole process of SPI. In addition, the proposed tool (named SysProVal) provides not only fundamental process assessment and improvement features, but also IP generation and remote online assessment and self-evaluation, respectively, depending on organisations.

3 Automated tool to support SPI initiatives in small settings

A frequent misconception about implementing CMMI and IDEAL is that they work only for large organisations (their cost and complexity appear to make it impractical for small businesses to implement). Nevertheless, there is a lot of research that contradict this assumption [14, 19, 20, 48–54]. We think that this idea may be supported if we are able to design and implement a web-based tool to manage the improvement information of any small organisation and guide it to improve its software process.

According to Jones [55], three major investment elements are involved in CMMIbased improvement:

- assessment,
- definition/infrastructure support, and
- deployment.

A process assessment involves a disciplined examination of the processes within an organisation. The SysProVal' assessment mechanism requires: A measurement scale – most commonly a series of maturity or capability levels; criteria for evaluation against the scale, usually an underlying maturity model; a set of standards, best practices and industry norms; and a clear mechanism for representing the results. The assessment model provides a common framework for assessing processes and forms a basis for comparison of results from different assessments. Many assessment techniques or methods are available: CBA-IPI, software capability evaluations, SCAMPI, interview-based, document intensive, questionnaire-based, mini-assessment. The assessment is used because it checks process improvement progress (against some standard), gains insight into program/project process, checks the maturity level and serves as an audit tool – to check a contractor/supplier in a procurement/contractual situation. We implemented a Questionnaire-Based Appraisal as an assessment tool because it can be applied to many people, it is cost effective and non-invasive, provides quantitative data, and the results can be analysed promptly [56].

3.1 Architecture

We are trying to provide a web-based tool that establishes an iterative approach to process improvement (PI) which a small organisation could adopt. Following this approach to PI, the focus of the first step would be to understand what exists in the organisation and determine what causes significant problems. Then solutions could be devised in the action plan and evaluated in pilot studies or even controlled experiments. Only after a solution is found to be effective and efficient, should it be integrated into the existing process or that process be modified. Such an approach is inspired by the Quality Improvement Paradigm recommended by several researchers and practitioners [57].

SysProVal' architecture is based on three layers separating the small setting logic, the assessment and improvement mechanisms and information retrieval (see Fig. 1).

This kind of architecture makes possible that through the logic layer, all SMEs take control over the complete SPI initiative and obtains, at the same time, customised results; all the business characteristics are evaluated with the assessment and improvement layer, establishing a real mapping of current software processes. The information retrieval layer manages all the generated information and uses the plan generator to send the complete information to clients. This architecture provides a light application for any client, and offers advantages related with implementation and management capabilities in a flexible computational program.

This architecture was developed to provide services for two kinds of users: project managers and top managers. Privileges and roles are restricted for each type of user.

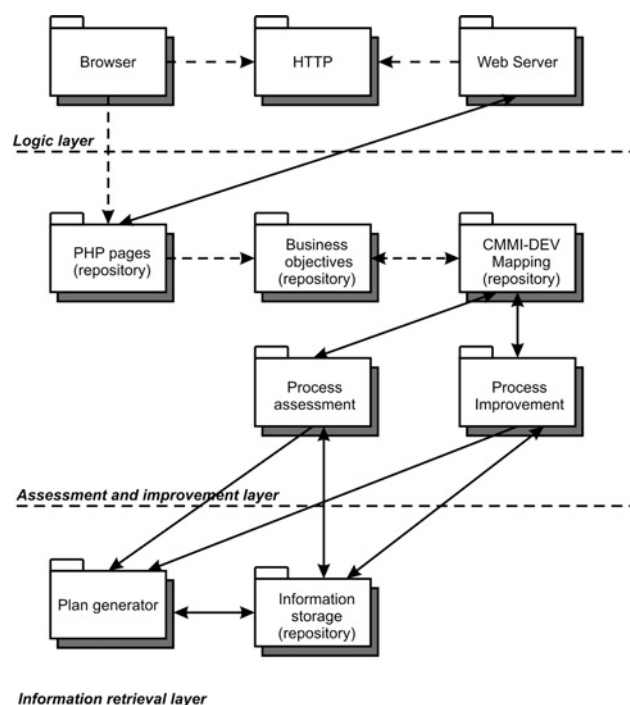


Figure 1 Architecture of SysProVal

A Top manager is responsible for controlling the project manager's performance in SysProVal and is the unique user with authorisation to manage the original configuration of the system; adding or eliminating processes for evaluating, selecting project managers for participating in the SPI initiative, controlling the recommendations, and managing and controlling all information in the organisation's repository. Fig. 2 shows the SysProVal's general diagram of use cases.

Fig. 3 illustrates an overall description for the SysProVal tool components. SysProVal consists of an SPI manager and mechanisms for: comparing the current practices with CMMI-DEV-adapted practices for SMEs; assessing the selected process; generating an adequate IP; and for using

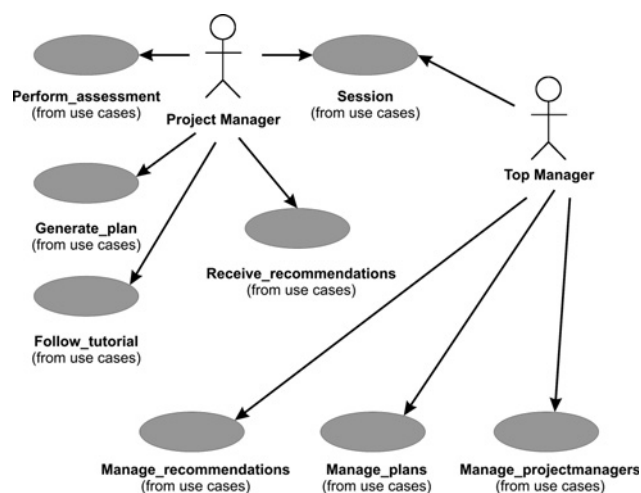


Figure 2 Diagram of use cases

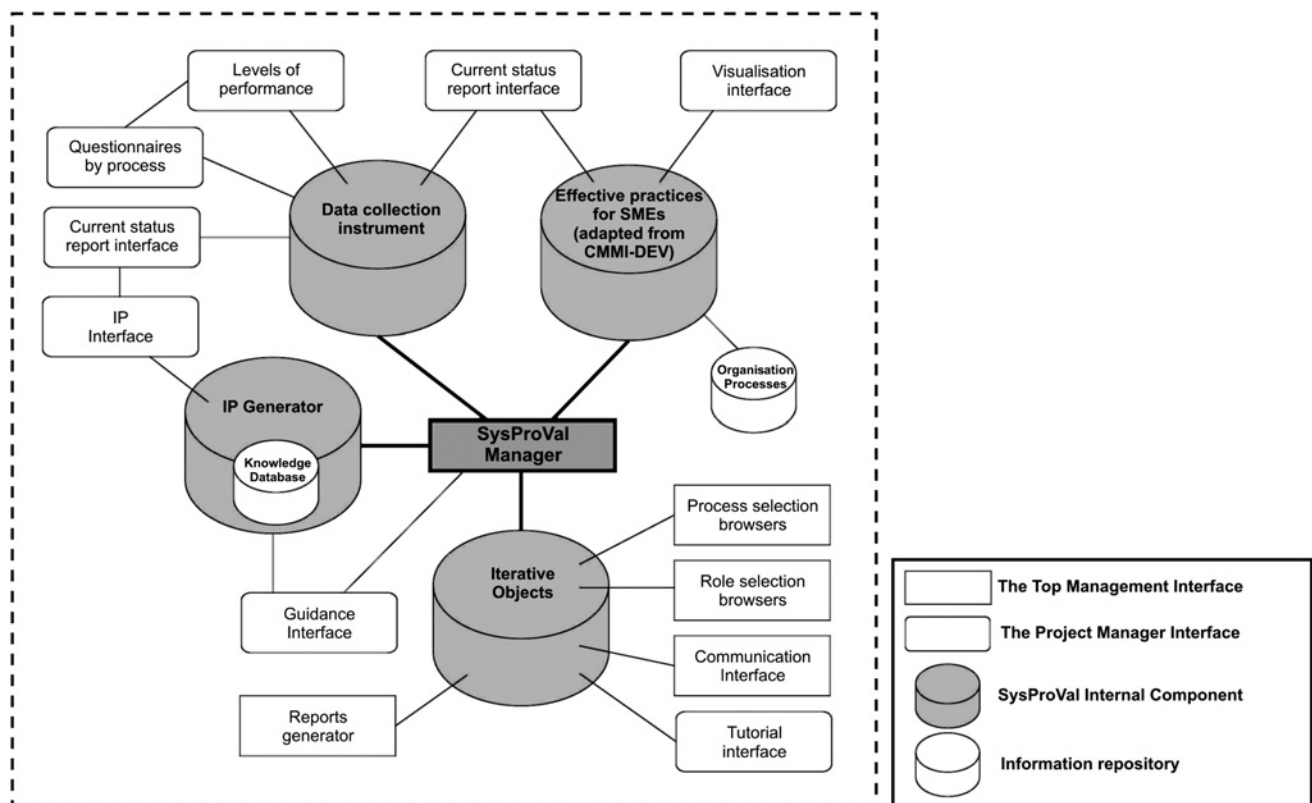


Figure 3 Components of SysProVal

the iterative elements of the tool. SysProVal provides three types of user interface components: (i) the top management interface, (ii) the project manager interface and (iii) the SysProVal interface.

3.2 Top management interface

The purpose of the top management interface is to support the whole range of SPI activities in an integral manner in terms of necessary guidance for SPI activities, communication support and learning support. The interface provides accesses for:

1. The process selection browser, which provides a checklist telling members of top management what processes can be selected. Each process selected is stored in the organisational processes repository and is associated with the description stored in the effective practices for SMEs component.
2. The role selection browser, which allows members of top management to select who will be evaluated from its project managers. They can also decide whether to delegate the internal role to the SysProVal manager, or address the SPI initiative with an external assessor.
3. The communication interface, which allows members of top management to send email messages using links incorporating objects from other interfaces, such as reports, questionnaires or new assignments. Through this interface,

top management can be informed about the progress of the SPI initiative, consulting results, reports or the progress status.

4. The reports generator, which allows members of top management to obtain any information on the SPI cycle at any time; they can control the performance of its project managers through the assessment process and obtain the final results and graphics derived from the entire process.

3.3 Project managers interface

The purpose of the project manager interface is to obtain all information about organisational practices to develop software products and to allow them to have a good understanding of what is going on in their development process (weaknesses and strengths).

1. The tutorial interface helps the project manager to obtain all knowledge needed to use the tool, interpret the results, understand each question and improve its work.
2. The visualisation interface provides the project manager with the tools to draw a new process or reuse existing ones (in accordance with CMMI-DEV activities).
3. The current status report interface presents a categorised level of performance, in accordance with the assessed process. The project manager can meet its own level of performance. This interface just provides performance level

results to project managers; the entire results of the organisation can only be reviewed by high-level management through the reports generator.

3.4 SysProVal interface

The purpose of the SysProVal interface is to allow top management of an SME to explicitly state the goals of the SPI activities, and to have a good grasp of the current progress of SPI activities in all improvement projects. Moreover, project managers can directly participate in the SPI activities through assessments and improvement plan (IP) generation.

1. The questionnaires interface guides project managers to obtain all knowledge from their daily labour. Previously, top management had selected which process to evaluate or what process to modify. This interface uses the 'levels of performance' from the two-phase mechanism proposed in [12] and resumed in Section 4.1.2.
2. The IP interface generates the IPs for each improvement project conducted. This interface uses knowledge management using databases to manage both information and people (see Section 4.1.3 for detailed information).

4 Implementing the IDEAL phases

4.1 Implementation

We have developed the proposed system in PHP using MySQL as a related database management system. PHP is free software released under the PHP License; we think that these conditions are favourable and affordable for any SMEs. This work is structured according to the first three phases of IDEAL [31] as we see it. Although these phases are presented sequentially, in most cases they are overlapping. The following phase's description uses a set of screenshots which correspond to a use scenario selected from one of the three pilot organisations that will be presented in Section 5.

4.1.1 Initiating: The SPI initiating phase was clearly covered and came from the executives of the organisations; therefore it was not necessary to get any buy-in or work to get executive support and commitment. According to Mondragon the PI infrastructure has three groups [11]: the management steering group (MSG), the engineering process group (EPG) and the technical working group (TWG). The membership for the MSG is drawn from senior executives. The EPG membership is drawn from respected project managers and practitioners. The recommended size of the group is 3% of the organisation's developers. The EPG ensures coordination of the PI effort throughout the organisation by providing process consultation, theme expertise, training and process improvement assessments. The TWG membership is drawn from line managers and practitioners from the

organisation areas affected by the process improvement project. TWG members assign approximately 20% of their time. For small settings, IDEAL recommends at least one full-time employee. Satisfying the one full-time employee requirement recommended in very small organisations is a challenge. The first question posed was from which part of the organisation should the PI leader be drawn? Should the PI leader be a top executive, a manager or a practitioner? We implemented the third approach (a practitioner as a leader of the PI effort) and it is currently providing the best results. This approach, however, requires full commitment from top management, support and respect from the managers of the PI leader, support from experts to document current and desired processes, written and oral communication skills from the PI leader, and the commitment from managers and practitioners to implement pilots. The second important issue in the initiating phase is the formation concept. We developed an SPI tutor that satisfies the knowledge acquisition process according to the SMEs necessities. This is not a certification course or even a formal course. This tutorial manages the lessons and promotes the inclusion of new topics according to the improvement project progress. If the performance of the project manager is low during the assessment, the tutorial recommends topics and alerts top management to its monitoring and control (see Figure 4).

4.1.2 Diagnostic: In order to characterise the initial state of the organisation's software processes as well as to assess the improvements done through the SPI cycle, we implement an assessment mechanism [23] inside SysProVal as a 'seek step' of PI efforts. This mechanism uses closed questions and limits the number of possible responses to seven. These are organised as follows:

- *Five level-perform-answers:* Always, Usually, Sometimes, Rarely if ever, and Never. These will enable us to know the extent to which each practice is performed.
- *Two validity-answers:* Don't Know and Does Not Apply. These will be used to appraise the validation of the questions, to validate how correct the question is, and to check the syntax of the questions.
- Additional information spaces (Comments) to extract supplementary background information. It is mandatory to write some comments when checking any of the validity answers.

Each possible response has a unique interpretation and indicates the performance level of a practice. The level-perform-answers determine the percentage in which each practice is performed. This varies from 'Never' with a value equal to 0, 'Rarely if ever' with a value equal to 1, 'Sometimes' with a value equal to 2, 'Usually' with a value equal to 3, and 'Always' with a value equal to 4. The validity-answers do not have numerical value. Giving a specific weight to each response will enable SysProVal to

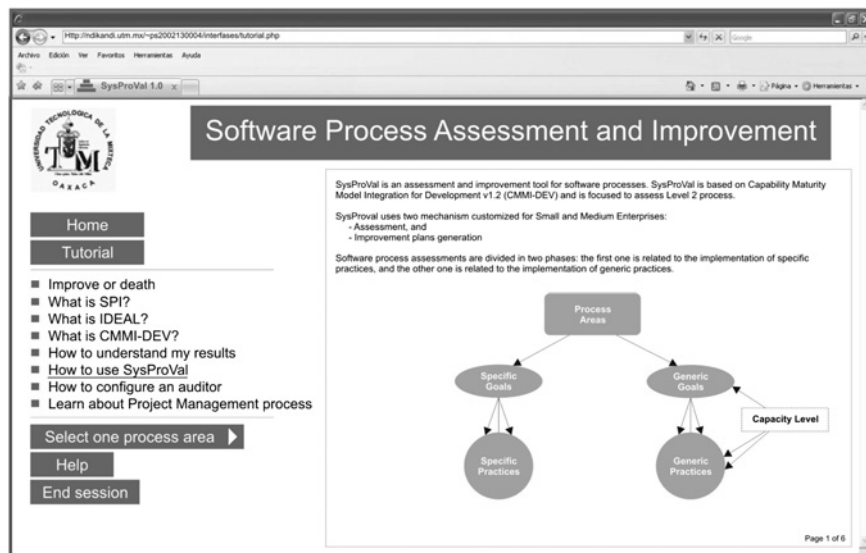


Figure 4 Screenshot of tutorial manager

easily analyse the results of the evaluation and to identify which practices are common within the whole organisation and which ones are not performed at all (see Fig. 5). The assessment mechanism proposed here is part of the questionnaires interface and has been based on the two types of practices established by the CMMIDEV dividing it into two phases. The first phase is related to specific practices and the second phase is related to generic practices.

As an example, the following figure shows the results of the initial SysProVal assessment for one small organisation through the current status report interface (see Fig. 6). The charts showed in Fig. 7 summarise the percentage of effective practices for SMEs component already addressed in the existing processes of the organisation, and the results are automatically grouped by SysProval according to CMMI-

DEV. This information is sent to top management by the reports generator as an 'improvement step'.

4.1.3 Establishing: The establishing phase begins with the 'improvement step', SysProVal sets the priorities, then develops the approach and finally generates the action plan to improve the current process with the obtained data from the previous step. The outcome of the 'improvement step' is a detailed implementation plan that is composed of 11 domains: specific actions, schedules, milestones, deliverables, decision points, resources, responsibilities, measurements, tracking mechanisms and risk management and mitigation strategies (see Fig. 8).

Clearly, the IP interface of the SysProVal tool, when applied to software projects in small organisations, produces

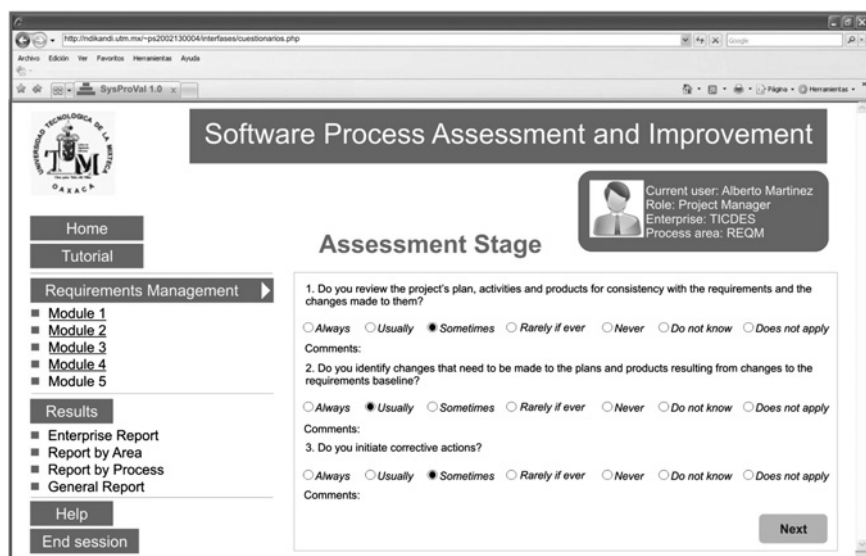


Figure 5 Screenshot of assessment interface

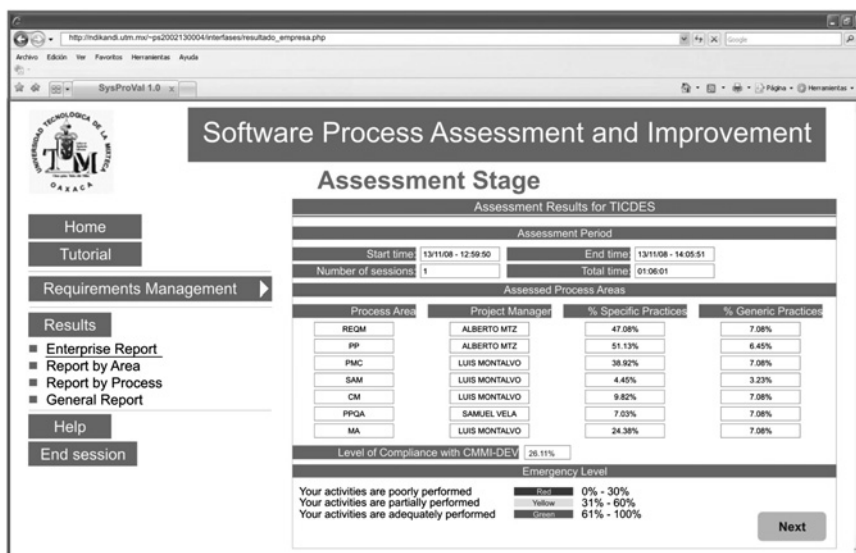


Figure 6 Screenshot of assessment results

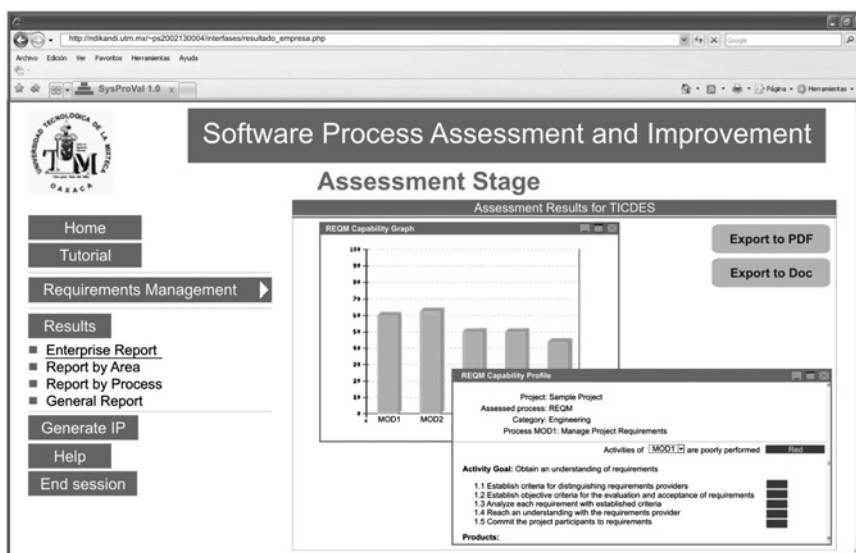


Figure 7 Screenshot of reports generator

a detailed plan with very similar features to those of big organisations. Therefore given the organisational experience of using CMMI models, it was natural to think of the CMMI-DEV as the supporting process for planning, tracking and controlling the SPI implementation activities for the improvement team.

The IP generator organised the obtained information from 'seek step' using databases. The most primary and preliminary requirement for a knowledge management in SysProVal is to manage the acquired data from the assessment interface. The contents are obtained through the recording of events and experiences in a specified format and capturing good and bad practices from web-based assessment. Conversion of performance levels in the form of information and storing

them in the database is achieved through the engineering technique of data mining. The transformation identifies important aspects of the knowledge domain, for example, notions, attributes, activities and associations with other process areas. This is similar to the management of 'what I know' in one's mind, storing what I have performed in the software environmental context and associating their activities with other successful activities (from CMMI-DEV). SysProVal identifies similar notions and these are clustered based on their attributes (poorly performed, partially performed and adequately performed). Giving a raw text to practice management section, the captured activities are represented in the form of 11 knowledge domains and are stored in an object-relational data model [58] of the organisation's processes.

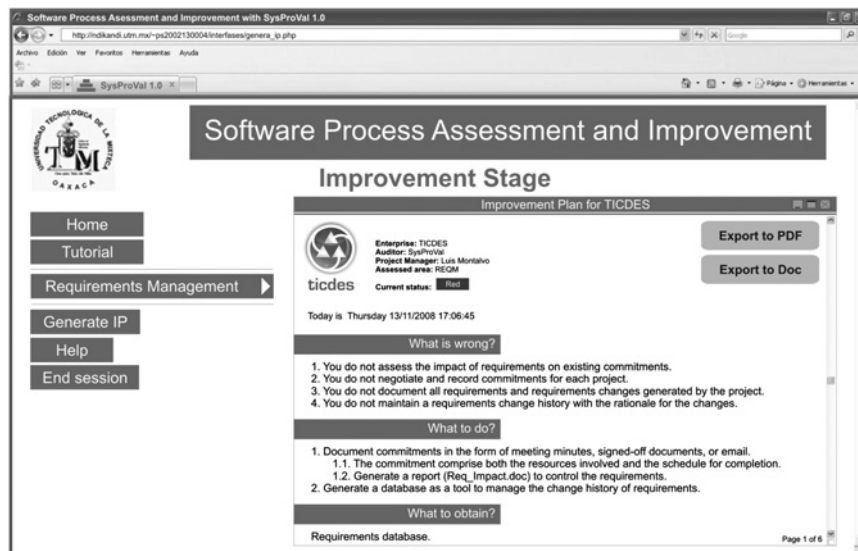


Figure 8 Screenshot of IP interface

5 Trial experience

The main objective of this research is to demonstrate that a small organisation may use a web-based tool (SysProVal) and the CMMI-DEV model as a framework to strengthen defined agile project management practices, improve the project performance and achieve high CMMI-DEV capability levels. We think that this idea may be applicable to other process areas, but the data that have been gathered so far corresponds to project management activities, thus we decided to limit the current scope research to the project management (PM) category of the CMMI-DEV model.

Our first goal is to articulate capability level 2 compliance with the defined web-tool discussed in this paper. This work is being carried out, and current findings will be presented here. Next, we will articulate capability levels 3 and 4. Theoretical capability levels 2 and 3 compliance will be documented by third quarter 2009, and actual capability level 2 compliance is expected to be confirmed during an official appraisal planned for one of the organisations in the first quarter of 2010.

project planning (PP) and project monitoring and control (PMC) were already analysed for capability level 2 compliance. We will continue to gather data for the implementations of these process areas and will articulate risk management and then integrated project management, organisational process definition and organisational process focus development to prepare for capability level 3. As a direct result of this project, information regarding the small settings community and the issues associated with software development were generated. This information, however, needed to be extracted, organised and analysed with SysProVal, in order to be useful in addressing the above research questions. The data consist of artefacts generated

as a result of the SysProVal evaluation. A high level overview of this evaluation is presented in Fig. 9.

The CMMI-DEV practices provide the environment and structure for this research. As part of this study these researchers created an advisory group for a small organisation to provide guidance for the project and insight into the environment of this organisation. With SysProVal, we provided the systems perspective important for the effectiveness of rapid assessment: 'Rapid appraisal should be based on what the participants in the system believe to be the critical elements, their relative importance and how they relate to each other' [59].

At the end of 2008 we conducted an empirical evaluation of SysProVal on small enterprises that provide solutions, consultancy services or who are involved in software development, plus many others. These enterprises were chosen for being typical of the Mexican Software Industry. For this study we primarily considered three small

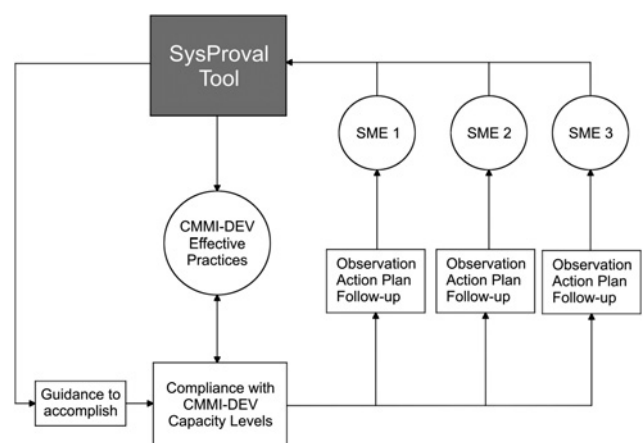


Figure 9 SysProVal evaluation setup

Table 2 Profiles of the enterprises

	Small enterprise		
	SME1	SME2	SME3
number of staff	20	15	17
deliverables	web-based portals	software solution providers	software solution providers and consultancy
SPI initiative	followed CMM standards, implementing effective practices	gathered past experiences from a knowledge repository	implemented SysProVal to participate in the SPI initiative
preliminary results	identified key process areas	framed a well defined and refined process set	identified two process areas for increasing quality and productivity

companies; we will name them SME1, SME2, SME3 for reasons of confidentiality. The former two are located in the state of Oaxaca and the latter in the state of Puebla. The nature of the survey is detailed in [Table 2](#).

The purpose of the evaluation was to demonstrate the ease and usefulness of SysProVal as an SPI tool for small enterprises, which also reduces the costs of undertaking an SPI initiative, considering the particular conditions of small settings.

We conducted an initial assessment with SysProVal to establish the base line capabilities of the enterprises processes according to CMMI-DEV v1.2. The result was 'poor' – between 0 and 1 (according to CMMI-DEV levels). During the next four months SysProVal coached the enterprises on SPI tailoring and adoption.

Finally, we applied the second assessment with SysProVal to each enterprise; all three enterprises achieved a 1.10 average increase in the capability level of two process areas: PP and PMC. [Table 3](#) shows the average capability level increase by process area.

[Table 4](#) shows the number of employees, total improvement effort in hours and effort per person for each enterprise. The last column includes the average capability improvement per process. It is interesting to note the relationship between the effort per person and the average process improvement when using SysProVal. The SME3 enterprise has invested the largest number of hours per person using SysProVal and achieved the greatest

Table 3 Capability level increase by category

Company	Process area	Increase
SME1	PP	0.90
	PMC	1.10
SME2	PP	1.10
	PMC	1.12
SME3	PP	1.21
	PMC	1.19
Global	1.10	

increment in its process improvement. The average number of employees was 17 and the average effort per person was 24.40 h over five months.

Commitment, initiative and responsibility were factors that strongly contributed to the SysProVal results obtained in this study. SysProVal was greatly concerned about communicating the importance of this SPI initiative, not only to the EPG, but to the entire team involved with the improvement programme. This raised the level of commitment and determination needed to achieve the future objective: the CMMI-DEV level 2. This empirical evaluation also enabled us to obtain a combination of valuable experiences, 'more knowledge on how to do it' and observation of behaviour during the process of improvement within small enterprises.

Table 4 Improvement and effort data by enterprise

Enterprise	# Employees	Total effort (h)	Effort per person (hours using SysProVal)	Average improvement
SME1	20	221	8.76	1.00
SME2	15	400	25.18	1.11
SME3	17	600	35.4	1.20
Average	17	407	24.40	1.10

5.1 Lessons learned and preliminary findings

We have reported on our early experiences in using a web-based tool as a foundation for implementing the CMMI-DEV in small organisations. Our research has shown that SysProVal makes the CMMI-DEV implementation easier through the first three phases of the IDEAL model. SysProVal has good coverage of these phases at a project level. However, it is important to ensure that the action plan (derived from CMMI-DEV) is followed faithfully by all the teams and project managers in the organisation. To this end, the 'seek step' mechanism proved to be a convenient and useful diagnostic tool.

As mentioned above, SysProVal has good coverage of the 'seek and improve' steps at team level. Therefore most of the CMMI-DEV implementation has to be at organisational level. Even if all the teams are following the CMMI-DEV by the book, there are still many organisational aspects that have to be covered.

Implementing an initiative to provide an SPI tool oriented to small settings and focused on the SEI family of products was a good decision. This family of products includes models, an implementation lifecycle model (IDEAL), assessment and appraisal methods. These products provide a consistent and complementary set of tools that facilitate the implementation of an SPI initiative. However, it is too difficult to adopt them in very small settings without a specialised guide. SysProVal is a huge shortcut to implementing such models. Actually, we are validating the acting and learning phases.

The effort and average improvement measures ensure us that the time destined to establish improvement initiatives can be reduced using SysProVal. However, there could be issues related with the human factor (out of our control) that were not contemplated in this experiment, but which could affect the obtained results. As a future work, we are incorporating psychometric techniques to improve the assessment process, which identify possible errors when SysProVal obtains information about the SME and as a consequence, provide more realistic IPs.

6 Conclusions

Although CMMI and ISO/IEC 15504 have exploded onto the market as models to follow when organisations want to apply process improvements, there are many organisations that are still not using these models. The CMMI is considered to be one of the best known models that focus on SPI for achieving quality software in small settings. The CMMI-DEV, however, is relatively new, so there has been little research written on which data collection and action plan generation tools can be employed when using the CMMI-DEV approach under the special conditions of small organisations.

This research, therefore, developed an instrument to define and implement SPI initiatives to improve the current status of project management practices. Its purpose was to investigate its feasibility in small settings and to influence the direction of future research. One limitation of this study is the generalisation of its findings based on the limited amount of data collected and analysed relative to the number of small organisations. This suggests that this qualitative study will be augmented by quantitative studies to strengthen the data supporting the need and applicability of SysProval to the small organisation community.

7 Acknowledgments

This paper is sponsored by ENDESA, Everis Consulting Foundation and Sun Microsystems companies through 'Research Group of Software Process Improvement in Latin America'.

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